

DLP Course: Basics of Geocomputation and Geoweb Services

Open Geodata Repositories & ISRO Geoweb Services for Thematic Applications

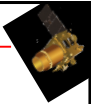


Kamal Pandey
Scientist/Engineer SE
kamal@iirs.gov.in

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- About open data repository (ODR).
- Types of data repositories.
- Modes of Data Access from ODR
- Key features of ODR
- Challenges in ODR
- Examples of Data Repositories from ISRO and other agencies
- Demo of Data Excess from ODR
 - OpenStreet Map
 - Bhuvan Geoportal

What is Open Geo-data Repository

- A data repository can be defined as a place that holds data, makes data available to **use, visualize and analyze**.
- A place where researchers can submit data.
- The repository which allows **receiving** and **reproducing** data from/by anyone is termed as **open data** repository



TYPES OF DATA REPOSITORIES

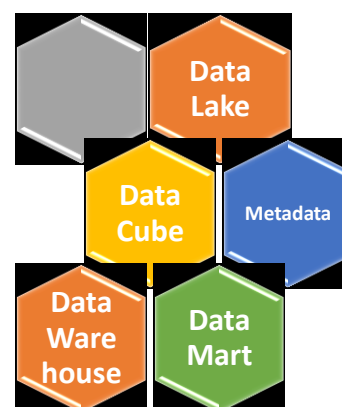
A **data warehouse** is a large data repository that aggregates data usually from multiple sources or segments of a business, without the data being necessarily related.

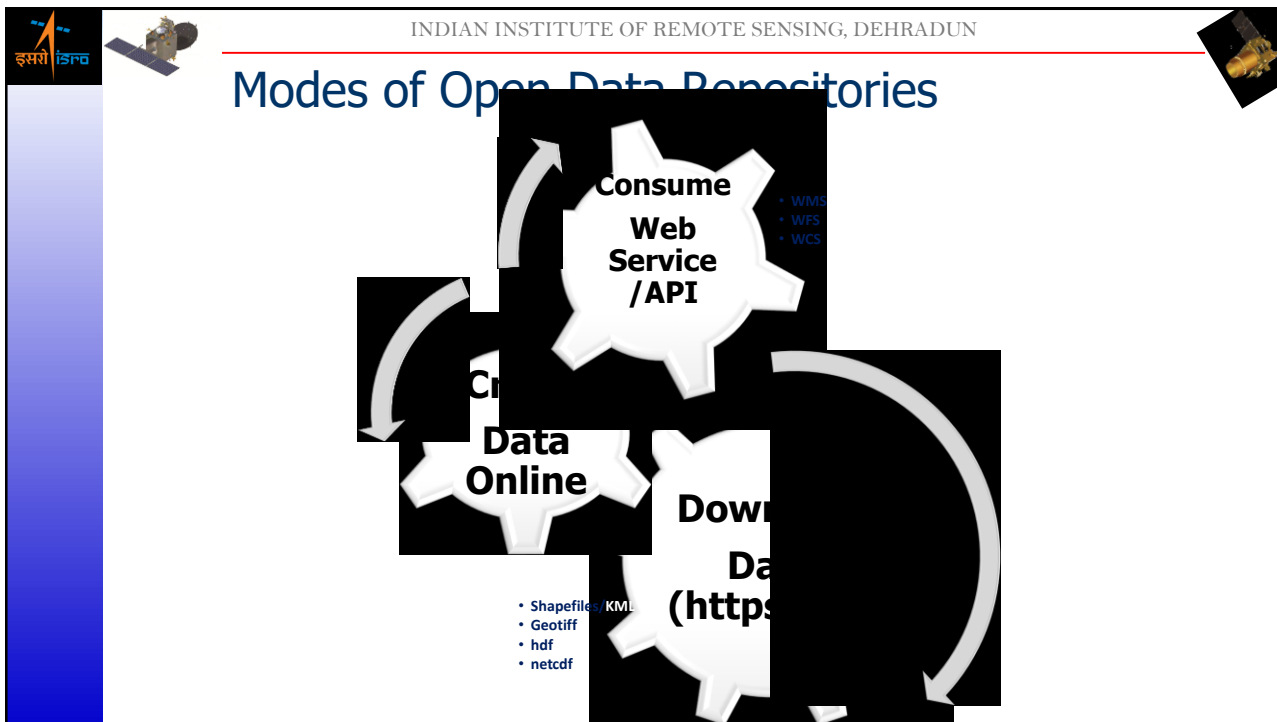
A **data lake** is a large data repository that stores unstructured data that is classified and tagged with metadata.

Data marts are subsets of the data repository. These data marts are more targeted to what the data user needs and easier to use. Data marts also are more secure because they limit authorized users to isolated data sets. Those users cannot access all the data in the data repository.

Metadata repositories store data about data and databases. The metadata explains where the data source, how it was captured, and what it represents.

Data cubes are lists of data with three or more dimensions stored as a table — as you may find in a spreadsheet





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Key features of Open Data Repositories

Availability and access

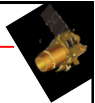
- The data must be available as a whole and at no more than a reasonable reproduction cost, preferably by downloading over the internet. The data must also be available in a convenient and modifiable form.

Reuse and redistribution

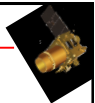
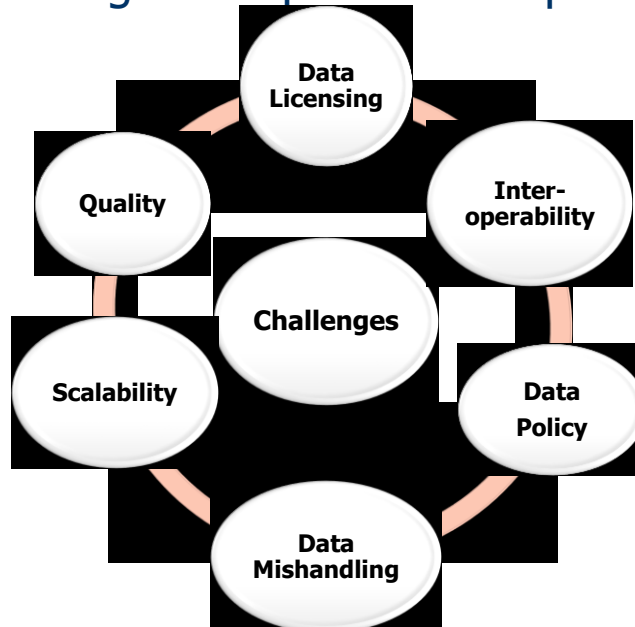
- The data must be provided under terms that permit reuse and redistribution including the intermixing with other datasets. The data must be machine-readable.

Universal participation

- Everyone must be able to use, reuse and redistribute — there should be no discrimination against fields of endeavour or against persons or groups. For example, 'non-commercial' restrictions that would prevent 'commercial' use, or restrictions of use for certain purposes (e.g. only in education), are not allowed.
- Open data can come from a variety of sources, typically government, but can also include data generated from research.



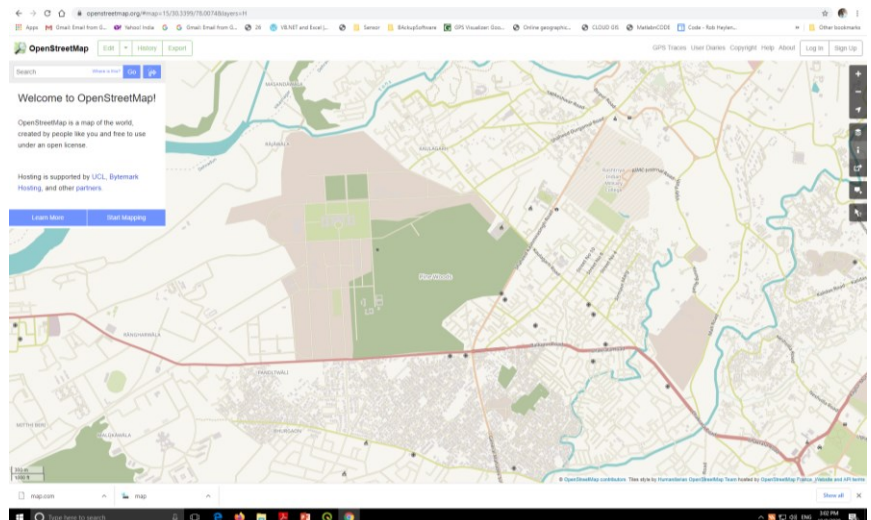
Challenges in Open Data Repositories



Examples Open Geo-Data Repositories

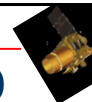
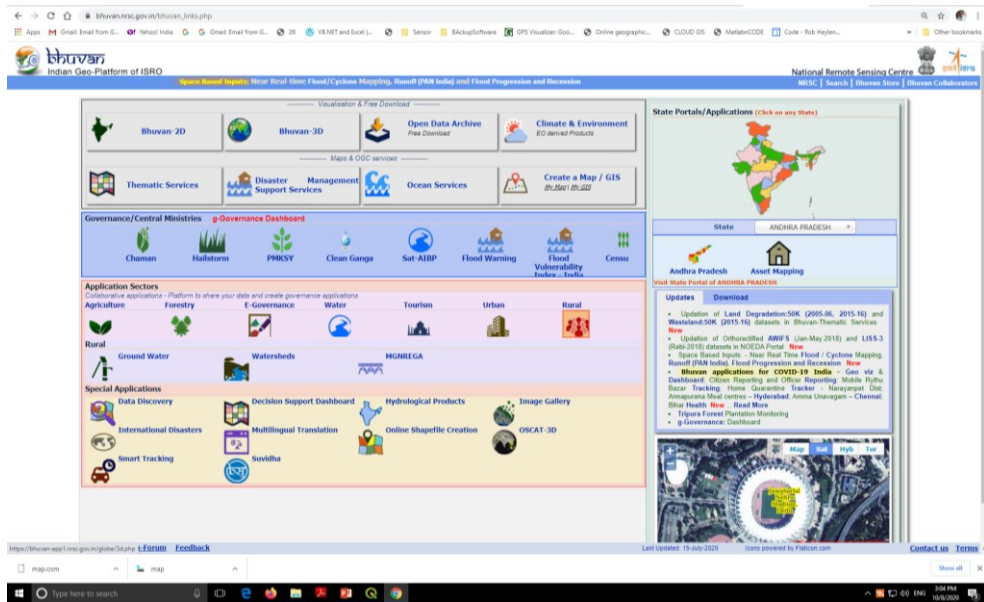
Demo: Download OSM Data

- From browser
- Using QGIS Software





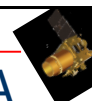
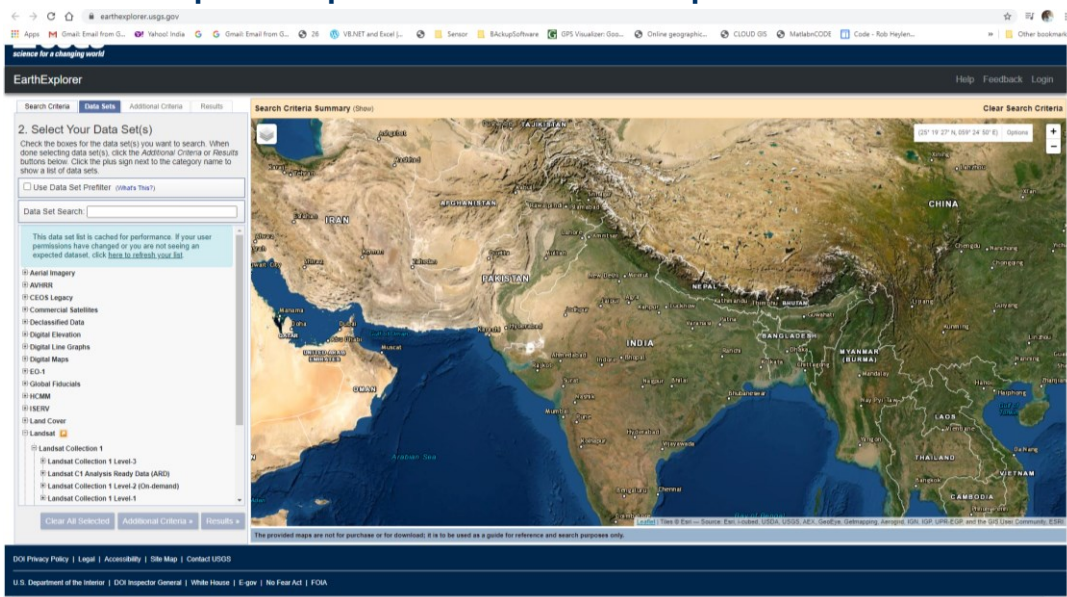
Examples Open Geo-Data Repositories: ISRO

The screenshot shows the Bhuvan Indian Geo-Platform of ISRO. The interface includes a search bar at the top, a navigation menu on the left, and a main content area with various data layers and maps. The navigation menu includes options like Bhuvan 2D, Bhuvan 3D, Open Data Archive, Climate & Environment, Thematic Services, Disaster Management Support Services, Ocean Services, and Create a Map / GIS. The main content area displays a map of India with various data layers, including Land Use/Land Cover, Forest, Water, and Urban. The right sidebar shows a list of updates and a map of India with a search bar.



Examples Open Geo-Data Repositories: NASA

The screenshot shows the EarthExplorer.usgs.gov interface. The interface includes a search bar at the top, a navigation menu on the left, and a main content area with a map of India. The navigation menu includes options like Search Criteria, Data Sets, Additional Criteria, and Results. The main content area displays a map of India with various data layers, including Land Use/Land Cover, Forest, Water, and Urban. The right sidebar shows a list of updates and a map of India with a search bar.

<https://www.geoseer.net>

Search Want a more intuitive way of spatially searching? Try the API Demos. About 4,400 results in 0.187s Village-Level Geospatial Socio-Economic Data for the state of Uttarakhand, India, 2001 nyu_2451_42009 http://maps-public.geo.nyu.edu/443/geoserver/sld/wfs nyu_2451_42009 Village Population c347 http://ds1.geoinfo.ac.th/geoserver/india/wfs c347 Quarantines c342 http://ds1.geoinfo.ac.th/geoserver/india/wfs c342 Village-Level Geospatial Socio-Economic Data for the state of Uttarakhand, India, 2001 nyu_2451_42009 http://maps-public.geo.nyu.edu/443/geoserver/ows?SERVICE=WMS& nyu_2451_42009 Village-Level Geospatial Socio-Economic Data for the state of Uttarakhand, India, 2001 nyu_2451_42009 http://maps-public.geo.nyu.edu/443/geoserver/ndv/ows?SERVICE=WMS& nyu_2451_42009 Village Population c347 http://ds1.geoinfo.ac.th/geoserver/india/ows?SERVICE=WMS& c347 1020 4.5 aip 1020_4_5_aip http://ds1.geoinfo.ac.th/geoserver/ows?SERVICE=WMS& 1020_4_5_aip	
WFS Uttarakhand, India, South Asia EPSG 4326	WFS Uttarakhand, India, South Asia EPSG 4326
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SOCIOECONOMIC DATA AND APPLICATIONS CENTER (SEDAC)

A Data Center in NASA's Earth Observing System Data and Information System (EOSDIS) — Hosted by [GESDIS](#) at Columbia University

 DATA
  MAPS
  THEMES
  RESOURCES
  SOCIAL MEDIA
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India Data Collection

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Collection Overview

Data Sets (3)

- [Spatial Data from the 2011 India Census, v1 \(2011\)](#)
- [Show All...](#)

[Map Gallery \(9\)](#)
[Map Services \(13\)](#)

Spatial Data from the 2011 India Census, v1 (2011)

[Set Overview](#)
[Data Download](#)
[Maps](#)
[Map Services](#)
[Documentation](#)
[Metadata](#)

Clicking on the map widget to the right will launch an interactive map tool which will allow you to zoom, pan and switch layers.

Here's an example on how to create a WMS layer instance using the Open Source Javascript library [OpenLayers](#).

```
var wms = new OpenLayers.Layer.WMS(
    "Population Density",
    "http://sedac.ciesm.columbia.edu/geoserver/tms/",
    {layers: "gsr:v1:gsr-v1-population-density-2011"}
);
```

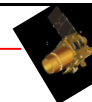
The possible values for `layers` can be found in the list below.

India Census 2011_Census Class	india-india-spatial-india-census-2011_census-class
India Census 2011_Census GHSL 1 Percent	india-india-spatial-india-census-2011_census-ghsl-1pct
India Census 2011_Census GHSL 50 Percent	india-india-spatial-india-census-2011_census-ghsl-50pct



2 of 3



Geoname

Example of Geoname API: (<https://www.geonames.org/export/web-services.html#countrycode>)

1. Search city based by pincode

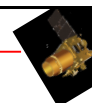
- JSON: <http://api.geonames.org/postalCodeSearchJSON?postalcode=248007&username=kamalisro>
- XML: <http://api.geonames.org/postalCodeSearch?postalcode=248001&maxRows=4&username=kamalisro>

2. Search Elevation based on latitude longitude information

- SRTM (90m): <http://api.geonames.org/srtm1?lat=30.2&lng=78.5&username=kamalisro>
- ASTER (30m): <http://api.geonames.org/astergdem?lat=30.2&lng=78.5&username=kamalisro>
- GTOPO (1KM): <http://api.geonames.org/gtopo30?lat=30.2&lng=78.5&username=kamalisro>

3. Search city based by Latitude Longitude information

- <http://api.geonames.org/findNearbyPlaceName?lat=30.35&lng=78.05&username=kamalisro>



Online free and open data repositories

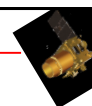
S.No	Product	URL	Coverage	Category
1	Gateway to Indian Earth Observation—Bhuvan Geportal	https://bhuvan.nrsc.gov.in/data/download/index.php	India	General and theme specific
2	Meteorological and Oceanographic Satellite Data Archival Centre— MOSDAC	https://www.mosdac.gov.in	India	General and theme specific
3	Indian Biodiversity Information System	http://bis.iirs.gov.in	India	General and theme specific
4	Indian Bio-resource Information Network IBIN	http://www.ibin.gov.in	India	General and theme specific
5	Visualization of Earth observation Data and Archival System (VEDAS)	https://vedas.sac.gov.in/vedas	India	General and theme specific
6	DIVA-GIS Country Data	http://www.diva-gis.org/gdata	Global	General
7	Natural Earth—Vector	http://www.naturalearthdata.com	Global	General
8	Global Map	https://www.iscgm.org/gmd	Global	General
9	UNEP GEOdata	http://geodata.grid.unep.ch	Global	General
10	Koordinates	https://koordinates.com/about	Global	General

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Online free and open data repositories				
S. No.	Product	URL	Coverage	Category
11	GDEM	http://asterweb.jpl.nasa.gov/gdem.asp	Global	Elevation
12	SRTM	http://srtm.csi.cgiar.org	Global	Elevation
13	Global Multi-Resolution Topography	http://www.marinegeo.org/portals/gmrt/	Global	Elevation
14	Open Topography	http://www.opentopography.org/index.php	Global	Elevation
15	WorldClim	http://www.worldclim.org	Global	Weather and climate
16	Climate Analysis Indicators Tool	http://cait.wri.org	Global	Weather and climate
17	HydroSHEDS	http://hydrosheds.cr.usgs.gov/index.php	Global	Hydrology
18	Delineation of major watersheds in the World	http://www.fao.org/geonetwork/srv/en/metadata.show?id=30914	Global	Hydrology
19	GeoNetwork	http://www.fao.org/geonetwork/srv/en/main_home	Global	General
20	GSHHS	http://www.ngdc.noaa.gov/mgg/shorelines/gshhs.html	Global	Land and Ocean Boundaries

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Online free and open data repositories				
S. No.	Product	URL	Coverage	Category
21	Water Isotopes	http://wateriso.utah.edu/waterisotopes/pages/data_access/data_main.html	Global	Hydrology
22	Natural Disaster Hazards	http://www.ideo.columbia.edu/chrr/research/hotspots/coredata.html	Global	Natural disasters
23	USGS Earthquakes Database	http://earthquake.usgs.gov/earthquakes/eqarchives/epic/kml	Global	Natural disasters
24	Global Seismic Hazard Map	http://www.seismo.ethz.ch/static/GSHAP	Global	Natural disasters
25	NOAA/WDC Historical Tsunami Database	http://www.ngdc.noaa.gov/hazard/tsu_db.shtml	Global	Natural disasters
26	MODIS Fire Detection Data	http://activefiremaps.fs.fed.us/gisdata.php	Global	Natural Disasters
27	Natural Disaster Hotspots	http://sedac.ciesin.columbia.edu/data/collection/ndh	Global	Natural disasters
28	GLOBCOVER	http://due.esrin.esa.int/page_globcover.php	Global	Land cover
29	MODIS Global Land Cover	http://visibleearth.nasa.gov/view.php?id=61004	Global	Land cover
30	EarthEnv Global Consensus Land Cover	http://www.earthenv.org/landcover	Global	Land cover



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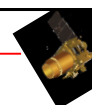


Online free and open data repositories

S. No.	Product	URL	Coverage	Category
31	Global Lakes and Wetlands Database	http://www.worldwildlife.org/pages/conservationscience-data-and-tools	Global	Land cover
32	World Soil Information	https://www.soilgrids.org/#/?layer=geonode:taxnrb_25_0m	Global	Ecology
33	Harmonized World Soil Database	http://webarchive.iiasa.ac.at/Research/LUC/External-World-soildatabase/HTML/index.html?sb=1	Global	Ecology
34	ERS/MetOp Soil Moisture	http://www.ipf.tuwien.ac.at/404/	Global	Ecology
35	Global High Resolution Soil Water Balance	http://www.cgiar-csi.org/data/global-highresolution-soil-waterbalance	Global	Ecology
36	Global Carbon Storage in Soils	http://www.docs_wri.org/page_forest_gh_sc_final.zip	Global	Ecology
37	World Bank Geodata	https://velluminformation.com/2012/01/28/google-earth-worldbank-data-and-kmlfiles/	Human geography	General
38	Google Maps Gallery	http://www.maps.google.com	Human geography	General
39	History Database of the Global Environment	http://themasites.pbl.nl/tridion/en/themasites/hyde/index.html	Human geography	General
40	Natural Earth	http://www.naturalearthdata.com	Human geography	Administrative boundaries



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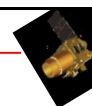


Online free and open data repositories

S. No.	Product	URL	Coverage	Category
41	IUCN 2013 Red List	http://www.iucnredlist.org/technicaldocuments/spatial-data	Human geography	Environmental boundaries
42	Global Land Use Dataset	http://nelson.wisc.edu/sage/data-and-models/global-land-use/grid.php	Human geography	Land use
43	Human Influence and Footprint	http://sedac.ciesin.columbia.edu/data/collection/wildareas-v2	Human geography	Land use
44	Crop Calendar GIS	http://nelson.wisc.edu/sage/data-and-models/crop-calendar-dataset/index.php	Human geography	Land use
45	ESPO Urban Morphological Data	http://database.espon.eu/db2/resource?idCat=43	Human geography	Land use
46	European Urban Morphological Zones	http://www.eea.europa.eu/data-and-maps/data/urban-morphologicalzones-2000-1	Human geography	Land use
47	Global Terrorism Database	http://www.start.umd.edu/gtd	Human geography	Wars, conflict and crime
48	Gridded Population of the World	http://sedac.ciesin.columbia.edu/data/collection/gpw-v4	Human geography	Population
49	Global Rural-Urban Mapping Project	http://sedac.ciesin.columbia.edu/data/collection/grump-v1	Human geography	Population
50	WorldPop	http://www.worldpop.org.uk	Human geography	Population



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Online free and open data repositories

S. No.	Product	URL	Coverage	Category
51	Large Urban Areas 1950-2050	https://nordpil.com/resources/worlddatabase-of-large-cities	Human geography	Population
52	Global Urban Extent	http://nelson.wisc.edu/sage/data-and-models/schneider.php	Human geography	Population
53	GeoHive	http://www.geohive.com	Human geography	Population
54	OpenStreetMap	http://www.geofabrik.de/data/download.html	Human geography	Buildings, roads and points of interest
55	OSM Metro Extracts	https://mapzen.com/data/metro-extracts	Human geography	Buildings, roads and points of interest
56	POI Factory	http://www.poi-factory.com	Human geography	Buildings, roads and points of interest
57	SimpleGeo's Places	http://s3.amazonaws.com/simplegeo-public/places_dump_20110628.zip	Human geography	Buildings, roads and points of interest
58	Nuclear Power Station locations	https://fusiontables.google.com/DataSource?dsrclid=579353#rows:id=1	Human geography	Buildings, roads and points of interest
59	Open Flights	http://openflights.org/data.html	Human geography	Transport and communications
60	World Port Index	http://msi.nga.mil/NGAPortal/MSI.portal?nfpb=true&pageLabel=msi_portal_page_62&pubCode=0015	Human geography	Transport and communications



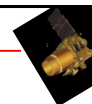
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S. No.	Product	URL	Coverage	Category
61	Global Roads Open Access Data Set	http://www.ciesin.columbia.edu/configuration/display/roads/Global+Roads+Data	Human geography	Transport and communications
62	Undersea Telecommunications Cables	http://www.cablemap.info	Human geography	Transport and communications
63	Geonames Country Information	http://download.geonames.org/export/dump/countryInfo.txt	Human geography	Gazetteers (place/feature names)
64	GRUMP Settlement Points	http://sedac.ciesin.columbia.edu/data/set/grump-v1-settlementpoints	Human geography	Gazetteers (place/feature names)
65	G-Econ	http://gecon.yale.edu	Human geography	Miscellaneous
66	Internet Map	http://www.netdimes.org/new/?q=node/65	Human Geography	Miscellaneous
67	World Urban Database and Access Portal Tools	http://www.wudapt.org/	Urban geography	Urban form and function
68	UN-Habitat Urban Data	http://urbandata.unhabitat.org/	Urban geography	Urban Indicators
69	Urban Development (World Bank)	http://data.worldbank.org/topic/urban-development	Urban geography	Urbanisation, traffic and congestion, and air pollution
70	DLR's Global Urban Footprint	https://urban-tep.eu/#/	Urban geography	Urban footprint



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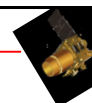


Popular Geoportals and their Services

S. No.	Web portal URL	Salient features
1	Google maps http://maps.google.com	Global satellite images and maps (2D and 3D) at street level with various applications
2	Bing Map http://www.bing.com/maps/	Global satellite images and maps (roads and POI)
3	Openstreet Map http://www.openstreetmap.org	Open and free street level vector data created using collaborative or participatory approach
4	Wikimapia http://wikimapia.org	Crowdsourcing approach for tagging ground Information
5	ISRO Bhuvan http://bhuvan.nrsc.gov.in	Rich contents and seamless availability of multi-temporal, multi-resolution satellite data for entire Indian region. Thematic and disaster services with data download facility
6	MapMyIndia http://www.maps.mapmyindia.com	Rich POI and detailed maps of India
7	Indian NSDI https://nsdiindia.gov.in	Metadata catalogue and policy document
8	Biodiversity information system http://bis.iirs.gov.in	Biodiversity spatial viewer and data download utility. Rich data contents on plant biodiversity of India
9	Indian Bio-resource information network http://ibin.gov.in	Rich data contents for Indian bio-resource information. System of distributed database
10	NIC GIS http://gis.nic.in	Village level tagging of information in GIS domain



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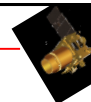


Thank You

Contact Details of the Faculty:

Phone: 01352524331

Email- kamal@iirs.gov.in



Other

- <https://population.un.org/wpp/Maps/>
- <http://data.un.org/Data.aspx?q=India&d=PopDiv&f=variableID%3a54%3bcrID%3a356>
- <https://libguides.tru.ca/opendata/repositories>

Example of Geoname API: (<https://www.geonames.org/export/web-services.html#countrycode>)

1. Search city based by pincode

- JSON: <http://api.geonames.org/postalCodeSearchJSON?postalcode=248007&username=kamalisro>
- XML: <http://api.geonames.org/postalCodeSearch?postalcode=248001&maxRows=4&username=kamalisro>

2. Search Elevation based on latitude longitude information

- SRTM (90m): <http://api.geonames.org/srtm1?lat=30.2&lng=78.5&username=kamalisro>
- ASTER (30m): <http://api.geonames.org/astergdem?lat=30.2&lng=78.5&username=kamalisro>
- GTOPO (1KM): <http://api.geonames.org/gtopo30?lat=30.2&lng=78.5&username=kamalisro>

3. Search city based by Latitude Longitude information

- <http://api.geonames.org/findNearbyPlaceName?lat=30.35&lng=78.05&username=kamalisro>